THERMAL PHYSICS LABORATORY:

CUBIC EXPANSIVITY OF A LIQUID AS A FUNCTION OF TEMPERATURE

The cubic expansivity of a substance is formally defined as the limit of the fractional change in volume that comes about due to some change in temperature at constant pressure as the temperature change becomes infinitesimally small (goes to zero):

$$\beta = \lim(\Delta T \to 0)\frac{\Delta V/V}{\Delta T} = \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_P$$

Thus, β could be determined experimentally by measuring the volume of a substance as a function of temperature.

In the experiment you will be doing something equivalent but slightly different: measuring the mass of a hot liquid displaced by a fixed volume as a function of temperture as the liquid cools. The fixed volume is the volume of a "Mathiessen sinker," which is a sealed glass bulb partially filled with lead shot such that it just sinks in water at room temperature. The mass of fluid displaced by the bulb is measured using Archimedes' Principle: as the fluid cools, its density increases, the mass of fluid displaced by the bulb increases, and the upward bouyant force increases.

Since you will be dealing with the temperature-dependent mass of the fluid displaced by the bulb, we need to re-write the expression for β in terms of changes in this mass. In the formula above, a fixed mass m_o of substance is assumed. Writing $V(T) = m_o/\rho(T)$, where $\rho(T)$ is the temperature-dependent density, we obtain

$$\begin{aligned}
\frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_P &= \frac{\rho}{m_o}\left(\frac{\partial}{\partial T}\frac{m_o}{\rho}\right)_P \\
&= \frac{\rho}{m_o}\left(-\frac{m_o}{\rho^2}\frac{\partial \rho}{\partial T}\right)_P \\
&= -\frac{1}{\rho}\left(\frac{\partial \rho}{\partial T}\right)_P \\
\beta &= -\left(\frac{\partial \ln \rho}{\partial T}\right)_P
\end{aligned}$$

The advantage of this formula is that it can be applied to the experimental setup you will be using, where, instead of a fixed mass occupying a variable volume, you have a variable mass of fluid occupying a fixed volume V_o:

$$\begin{aligned}
m(T) &= \rho(T)V_o \\
\ln m(T) &= \ln \rho(T) + \ln V_o \\
\beta &= -\left(\frac{\partial \ln \rho}{\partial T}\right)_P = -\left(\frac{\partial \ln m}{\partial T}\right)_P
\end{aligned}$$

EXPERIMENTAL PROCEDURE

1. Ensure that the unloaded triple beam balance reads zero. Weigh the Mathiessen sinker in air. You should strive to obtain all mass measurements in this experiment with accuracy to the smallest sub-division on the 0-1 gm scale, i.e. a hundredth of a gram.

2. Heat the water to about 75° C and *very carefully* transfer the beaker and water to the support stand on the balance.

3. Carefully lower the bulb, suspended by a thread from the balance, into the beaker of hot water. Ensure that the sinker is fully submerged. Weigh the bulb again, ensuring that the beaker is not in contact with the balance, and record the corresponding temperature of the water.

4. Record the temperature and the mass of the fully-submerged sinker as the water cools to tap-water temperature. Initially, one reading should be taken for every 0.1 gm mass decrease, which should correspond to two or three degrees of temperature decrease. As the temperature becomes lower, you will notice a steady growth in the temperature drop per 0.1 gm increase in the sinker mass. When this temperature drop exceeds five degrees, reduce the mass increment between readings to 0.05 gm.

   Note: A convenient way to make the mass and temperature measurements is as follows. After a measurement, adjust the rider on the 1-gm beam so that it is sitting on the next mass increment you want to measure; initially, the sinker will be somewhat more massive than the water it displaces (i.e. the beam pointer will be above the balanced position). As the water cools, the beam pointer will come down toward the balanced position - you can then wait for the beam pointer to pass through the balanced position and read the temperature corresponding to this time (you don't have to scramble to read the mass, because you specified it just after the previous measurement). Then adjust the rider on the 1-gm beam so that it is sitting on the next mass you want to measure and carry on in this fashion.

5. Initially, the cooling rate will be such that prompt recording of temperature and mass will be required in order to achieve a good density of data points. The cooling rate will gradually slow, since it is proportional to the constantly decreasing difference between the temperatures of the water and air. The latter stages of the cooling process can be accelerated by removing some water from the beaker and replacing it with tap water using the equipment provided. *But, if you do this, be careful not to disturb the balance, and ensure that any drops of water that attach to the balance or its wire frame are dried off!*

Setting Up

You are free to use any software package you wish to do this data analysis.

1. Enter your temperature (in degrees Celsius) and mass readings as two columns of data in a spreadsheet.
2. Add another column of data with the mass of the water displaced, obtained by subtracting each mass reading from the mass of the bulb in air.
3. Add another column of data with $\ln m(T)$.

General Examination of the Data

1. Make a plot of your points for $\ln m(T)$ vs. temperature for your entire dataset.
2. Obtain a linear fit to the data:

$$\ln m(T) \;=\; a + bT$$

   and verify that such a fit does not provide a fully adequate description of the data points by plotting the points and the fit.

3. Now obtain a quadratic fit to the data:

$$\ln m(T) \;=\; a + bT + cT^2$$

   Does this seem to provide a good fit to the data – does it seem to reproduce well all the bumps and dips in the temperature dependence for $\ln m(T)$?

4. Now obtain a cubic polynomial fit to the data:

$$\ln m(T) \;=\; a + bT + cT^2 + dT^3$$

   Is the fit provided by this function visibly better than the fit provided by the quadratic fit?

5. Find the root-mean-square differences between your N data points and the quadratic and cubic polynomial fits $f(T)$, together with the corresponding rms percentage errors:

$$\sqrt{\frac{1}{N}\sum_{i=1}^{N}(\mathrm{Data}_i - f(T_i))^2} \qquad\qquad \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(\frac{\mathrm{Data}_i - f(T_i)}{\mathrm{Data}_i}\right)^2}$$

   Do these quantities for the two different polynomial fits indicate that one fit is significantly better than the other?

Determination of the Cubic Expansivity

1. Determine a polynomial approximation for β as a function of the temperature by taking the negative derivative of the cubic polynomial fit used to approximated $\ln m(T)$ as a function of T:

$$\beta = -b - 2cT - 3dT^2$$

2. The most precise experimental determination of the cubic expansivity of water at 1 atm pressure is very accurately interpolated by the function

$$\beta = 10^{-6}\frac{-68 + 16.9T}{1 + 0.01629T - 4.68 \times 10^{-5}T^2}$$

where T is likewise in degrees Celsius – we will take this to be a mathematical expression for the "true" value of β. Make a plot of the expression you got in step 1 just above together with a plot of this function – how well do the two functions agree? If the two functions don't have roughly the same magnitude, check – you may have made a mistake somewhere.

3. Make a plot of the difference between these two functions for each of the temperatures you measured. Do the two functions agree equally well throughout the entire temperature range, or are there certain intervals of temperature where they don't agree as well? If the latter, do you have any ideas about why this might be the case?

4. Find a value for β for water for atmospheric pressure and some temperature in your range of measured temperatures in a handbook or other reference book – how does this value compare with the value indicated by your data for that temperature?